

MACHINE MANUAL

CUSTOMER QUESTIONNAIRE

Comprehensive Data Collection Form
Operations | Maintenance | Troubleshooting

Version 1.0 | June 2026

CONFIDENTIAL

HOW TO USE THIS QUESTIONNAIRE

This questionnaire captures machine-specific operational, maintenance, and troubleshooting data required to develop your Operating & Safety Manual.

Fill in ALL fields. If information is unavailable, write "Not Available" or "Unknown". Do not leave any field blank.

Safety-critical data (LOTO, E-Stop, Guards, Hazards) is sourced separately from the Risk Assessment Report and is not included here.

For file uploads (schematics, drawings, alarm exports), attach as separate documents and reference the item number below.

Tip: Sections F and G (Maintenance and Troubleshooting) are the most important. Spend maximum time here.

PROJECT INFORMATION

Project Code	
Machine Name	

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Customer / Plant	
Site Address	
Date of Visit	
Completed by	
Designation	
Contact Number	

PART A — MACHINE IDENTITY

A1. Machine name

Exact name as shown on the nameplate

Response: _____

A2. Machine type / category

e.g., Assembly machine, Drilling machine, Test bench

Response: _____

A3. Manufacturer / OEM

Company name as on nameplate

Response: _____

A4. End user / plant

Site name and full address

Response: _____

A5. Model number

As per nameplate

Response: _____

A6. Serial number

As per nameplate

Response: _____

A7. Year of manufacture

YYYY

Response: _____

A8. Installation location

Building / line / bay / bay coordinates

Response: _____

A9. Main contact — Engineering

Name, role, email, phone

Response: _____

A10. Main contact — Production

Name, role, email, phone

Response: _____

A11. Main contact — Maintenance

Name, role, email, phone

Response: _____

A12. Main contact — EHS

Name, role, email, phone

Response: _____

PART B — MACHINE PURPOSE AND PROCESS

B1. What does the machine do?

Describe in one clear sentence — product made or task performed

Response: _____

B2. Product / component processed

Part name or part number(s) if applicable

Response: _____

B3. Intended use

Describe what the machine is designed to do and its limits

Response: _____

B4. Prohibited use / foreseeable misuse

List actions or uses that are NOT permitted

Response: _____

B5. Input to machine

Material, part, tray, or fixture loaded into the machine

Response: _____

B6. Output from machine

Finished product, OK / NG routing, output location

Response: _____

B7. Process sequence — stepwise

Step 1 → Step 2 → Step 3 ... (describe each station/action)

Response: _____

B8. Station names and functions

List each station by name and describe its primary function

Response: _____

B9. Normal cycle time

Seconds per piece, or seconds per cycle

Response: _____

B10. Rated throughput

Units per hour (UPH) / parts per hour (PPH) / tonnes per minute (TPM)

Response: _____

B11. Operator count and roles

Number of operators and what each role does

Response: _____

B12. Operator manual actions during cycle

Describe every task the operator physically performs during normal operation

Response: _____

B13. Material movement types

e.g., Conveyor, robot, gantry, manual fixture, gravity feed

Response: _____

PART C — OPERATING PROCEDURES

C1. Pre-start checks list

List every check the operator must complete before pressing Start

Response: _____

C2. Utilities required

Electrical, compressed air, hydraulic, cooling water, vacuum — specify each

Response: _____

C3. Exact startup sequence

Step-by-step operator actions from main power ON to machine ready

Response: _____

C4. Normal operating sequence

Step-by-step description of one complete production cycle

Response: _____

C5. Conditions that stop the machine automatically

List all sensors, alarms, or interlocks that trigger an auto-stop

Response: _____

C6. Normal shutdown sequence

Step-by-step operator actions from end of production to machine fully off

Response: _____

C7. Emergency shutdown — operator action

Exact steps operator must take in an emergency (NOT the LOTO procedure)

Response: _____

C8. Power failure recovery procedure

Actions required after unexpected power loss and power restoration

Response: _____

C9. E-stop reset procedure

Exact steps to reset and verify the machine after an E-stop

Response: _____

C10. Operating modes and access permissions

List all modes (Auto / Manual / Setup / Jog etc.) and who is authorised for each

Response: _____

PART D — HMI / IPC / ALARMS

D1. HMI / IPC make, model and software version

e.g., Siemens TP1200 Comfort, WinCC V16

Response: _____

D2. HMI screenshots available?

Attach screenshots of all main screens as image files

☐ Yes — will attach ☐ No

D3. Alarm list export available?

CSV export from PLC / HMI / SCADA is preferred

☐ Yes — CSV attached ☐ Yes — manual list provided ☐ No

D4. Alarm details — for each alarm provide:

Alarm code | Alarm text | Severity (High/Med/Low) | Probable cause | Operator action | Maintenance action | Reset procedure

Response: _____

D5. Are alarms logged / recorded?

If Yes, specify where — PLC memory / SCADA / MES / SQL database

☐ Yes ☐ No

PART E — OPERATOR TASKS

E1. Operator load / unload tasks

Describe exactly what the operator loads and unloads, and at what frequency

Response: _____

E2. Consumables replenished by operator

e.g., fasteners, lubricant, cutting fluid, filters — specify replenishment frequency

Response: _____

E3. Operator visual and quality checks

What does the operator inspect? What are the pass/fail acceptance criteria?

Response: _____

E4. Records maintained by operator

Logbook entries, production counts, fault records — specify format and frequency

Response: _____

E5. Faults the operator is permitted to clear

List specific alarm codes or fault conditions the operator may reset independently

Response: _____

PART F — MAINTENANCE TASKS

This is the most critical section. Take maximum time here. If a PM schedule exists, attach it under J6.

F1. Maintenance roles and authorisations

Who performs maintenance (designation) and what is each role authorised to do

Response: _____

F2. Daily maintenance tasks

List each task with its acceptance / pass criteria

Response: _____

F3. Weekly maintenance tasks

Response: _____

F4. Monthly maintenance tasks

Response: _____

F5. Quarterly maintenance tasks

Response: _____

F6. Annual maintenance tasks

Response: _____

F7. Tasks requiring full LOTO before starting

List which maintenance tasks require machine to be fully de-energised and locked out

Response: _____

F8. Lubrication points

For each point: Component name | Lubricant type & grade | Quantity | Method | Interval

Response: _____

F9. Wear parts and replacement intervals

Part name / number | Expected service life or interval | Replacement method

Response: _____

F10. Calibration tasks

What is calibrated | Frequency | Acceptance criteria / tolerance

Response: _____

F11. Spare parts list / BOM available?

Attach as CSV or Excel under J7

☐ Yes — will attach ☐ No

PART G — TROUBLESHOOTING

Focus on real faults that have actually occurred on this machine. Generic faults are less useful.

G1. Top common faults — list with symptoms

Describe the top 5–10 faults: what the operator sees, hears, or detects

Response: _____

G2. Fault details — for each fault provide:

Symptom | Root cause | Operator action | Maintenance action | Reset / recovery procedure

Response: _____

G3. Conditions requiring immediate machine shutdown

List every condition that must result in the operator stopping the machine immediately

Response: _____

PART H — UTILITIES AND TECHNICAL DATA

H1. Electrical supply specification

Voltage (V) | Phases | Frequency (Hz) | Incoming breaker rating (A)

Response: _____

H2. Control circuit voltage

e.g., 24 V DC, 110 V AC

Response: _____

H3. Total connected load

kW — as per electrical panel or nameplate

Response: _____

H4. Pneumatic specification

Supply pressure (bar) | Air quality class | Fitting size

Response: _____

H5. Hydraulic specification

Hydraulic fluid type & grade | System pressure (bar) | Tank capacity (litres) — enter N/A if not applicable

Response: _____

H6. Cooling water / chiller specification

Flow rate (LPM) | Inlet temperature (°C) | Return temperature (°C) — enter N/A if not applicable

Response: _____

H7. Environmental operating limits

Ambient temperature range (°C) | Humidity (%) | Dust or contamination class

Response: _____

H8. Machine dimensions and weight

Length × Width × Height (mm) | Total weight (kg) | Attach GA drawing under J2 if available

Response: _____

PART I — SAFETY AND ACCESS

Note: LOTO procedures, E-Stop layout, guard details, and hazard identification are captured from the SICK Risk Assessment report — not in this form.

I1. Authorised personnel categories

List all categories: e.g., Operator / Setup Technician / Maintenance Technician / Safety Authorised Person — describe each role

Response: _____

12. PPE required — by task

For each task or area, list the required PPE: e.g., Safety shoes (all areas), Cut-resistant gloves (blade change), Face shield (coolant flush)

Response: _____

13. Permits required for maintenance work

e.g., Hot work permit, Confined space entry permit, Working at height permit — specify when each applies

Response: _____

PART J — REQUIRED ATTACHMENTS

For each item below, indicate status and attach the file (or provide folder / share link).

J1. SICK Risk Assessment Report

Mandatory where available — primary source for safety sections

☐ Attached ☐ Will send later ☐ Not available

J2. GA drawing / Layout drawing

☐ Attached ☐ Will send later ☐ Not available

J3. Electrical schematic / wiring diagram

☐ Attached ☐ Will send later ☐ Not available

J4. Pneumatic schematic

☐ Attached ☐ Will send later ☐ Not available

J5. Hydraulic schematic

Enter N/A if machine has no hydraulic system

☐ Attached ☐ Will send later ☐ Not available

J6. Maintenance checklist or PM schedule

☐ Attached ☐ Will send later ☐ Not available

J7. Spare parts list / BOM

CSV or Excel format preferred

☐ Attached ☐ Will send later ☐ Not available

J8. HMI alarm export

CSV preferred — can be exported from PLC or SCADA

☐ Attached ☐ Will send later ☐ Not available

Attachment folder / share link (if not emailing files individually):

Response: _____

PART L — OPERATOR INTERVIEWS

These questions must be asked directly to experienced machine operators — not the supervisor or engineer. Answers capture tacit field knowledge that no manual or drawing will contain.

Ask at least 2 operators independently and compare answers. Discrepancies reveal undocumented variations in practice.

Operator Details

L0a. Operator name, designation, years on this machine

Response: _____

Interview Questions

Q1: What is the one thing that goes wrong most often on this machine — and what do you do when it happens?

Answer: _____

Q2: Is there anything you do differently from what you were trained to do? If yes, why?

Answer: _____

Q3: What does this machine sound or feel like just before something goes wrong? Any early warning signs?

Answer: _____

Q4: Which parts of this machine wear out fastest or need replacing most often?

Answer: _____

Q5: What is the most dangerous moment or action when working on or near this machine?

Answer: _____

Q6: If you were training a new operator today, what is the one thing you would absolutely make sure they know?

Answer: _____

Additional Notes from Observation

Response: _____

PART M — PROJECT REQUIREMENTS

M1. Manual primary use

e.g., Operator training / Safety compliance / Customer audit / Regulatory submission

Response: _____

M2. Number of copies required

Printed copies + digital (PDF) copies

Response: _____

M3. Language(s) required

e.g., English only / English + Tamil / English + Hindi

Response: _____

M4. Target delivery date

DD-MMM-YYYY

Response: _____

M5. Any specific sections or topics to prioritise

Response: _____

M6. Additional notes or special requirements

Response: _____

PART K — DECLARATION

For every field in this questionnaire where information could not be provided, I confirm that the response "Not Available" or "Unknown" has been entered, and no field has been left blank.

K1. Data completeness declaration

Tick to confirm

☐ Confirmed — all fields answered

K2. Accuracy agreement

Tick to confirm

☐ I confirm that the data provided is accurate to the best of my knowledge and no values have been invented or assumed

SIGN-OFF

Consultant / Prepared by: _____ Date: _____

Customer Representative: _____ Role: _____

Signature: _____ Date: _____